

Literature Status: Removing the HCR Hypothesis for Cyclic Reproducing Kernels in $\mathcal{H}(b)$

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Status: literature already answered. Question 5.2 of arXiv:2210.15735 asks whether the HCR/corona-pair hypothesis in its Proposition 5.7 can be removed. Theorem 4.2, Corollary 5.1, and Example 5.2 of the separate later paper arXiv:2403.04335 answer both parts affirmatively. The later authors explicitly state that the earlier corona-pair assumption can be omitted.

1 Original question

Let b be a non-extreme point of the closed unit ball of H^∞ , let a be its Pythagorean mate, and let S_b denote the forward shift on the de Branges–Rovnyak space $\mathcal{H}(b)$. Fricain and Grivaux prove in Proposition 5.7 of [1], under their condition (HCR), that every reproducing kernel

$$k_\lambda^b(z) = \frac{1 - \overline{b(\lambda)}b(z)}{1 - \overline{\lambda}z}$$

is cyclic for S_b , and that if b is outer then bk_λ is cyclic for every $\lambda \in \mathbb{D}$ (hence b itself is cyclic).

On arXiv PDF page 20, the paper ends with:

Source Question 5.2. *Does Proposition 5.7 hold without the assumption that (a, b) satisfies (HCR)?*

2 Separate later answer

Fricain and Lebreton prove the following sufficient condition in Theorem 4.2 of [2]: if $f \in \mathcal{H}(b)$ is outer and $b/f \in L^\infty(\mathbb{T})$, then f is cyclic for S_b . Their Corollary 5.1 gives two immediate consequences:

1. if $\inf_{\mathbb{D}} |f| > 0$, then f is cyclic;
2. b is cyclic if and only if b is outer.

Example 5.2 on arXiv PDF page 16 applies these statements to the exact two parts of Proposition 5.7. For each $\lambda \in \mathbb{D}$,

$$|k_\lambda^b(z)| = \left| \frac{1 - \overline{b(\lambda)}b(z)}{1 - \overline{\lambda}z} \right| \geq \frac{1 - |b(\lambda)|}{2} > 0,$$

so Corollary 5.1(i) makes k_λ^b cyclic. If b is outer, then bk_λ is outer and

$$\frac{b}{bk_\lambda} = \frac{1}{k_\lambda} = 1 - \bar{\lambda}z \in L^\infty(\mathbb{T}),$$

so Theorem 4.2 makes bk_λ cyclic. Corollary 5.1(ii) also gives the stated cyclicity of b itself.

The authors then explicitly note that these were the two results previously obtained in Proposition 5.7 under the corona-pair assumption, and conclude that the assumption can be omitted. Thus Question 5.2 has a complete affirmative answer.

3 Scope

The identification is exact: it covers every non-extreme b , every $\lambda \in \mathbb{D}$, and both conclusions of the source proposition. No new mathematical result is claimed in this packet.

References

- [1] E. Fricain and S. Grivaux, *Cyclicity in de Branges–Rovnyak spaces*, arXiv:2210.15735.
- [2] E. Fricain and R. Lebreton, *Cyclicity of the shift operator and a related completeness problem in de Branges–Rovnyak spaces*, arXiv:2403.04335.